

Maintaining Reliability Under Extreme Conditions in Space

The electronics on-board space exploration vehicles should be able to withstand the extreme temperatures of space. Operating it beyond the normal limits is an option worth considering



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Space exploration vehicles like satellites, probes, shuttles and spacecrafts all rely on electronics. These electronic devices have to withstand for their working not only the extreme (very low to very high) temperatures but also the

radiation hazards prevailing in the space. In fact, very little of a space system falls within the conventional electronics' temperature range. Its temperature in space depends on its proximity and orientation to other bodies, absorption and emission of energy, and internal heat generation. Therefore extreme-temperature elec-

tronics is a key technology for space exploration.

Use of conventional electronics

Conventional-temperature-range electronics can be used in space (an extreme-temperature environment) by means of insulation and heating (for low-temperature environments) or refrigeration (for high-temperature environments). This can be combined with thermal sinks or thermal sources. For example, the well-logging electronic devices may be placed in a dewar flask (vacuum-insulated vessel) to protect these from the hot environment.

In addition, a thermal sink thermally connected to the electronic devices will absorb a large amount of heat without a substantial increase in temperature. This is usually done by employing the material's phase change from solid to liquid, which absorbs a large amount of heat (the latent heat of fusion). For low temperatures, the opposite effect may be used to provide a thermal source. Thus the same material may serve both as a thermal sink for high temperatures and a thermal source for low temperatures. It might be as basic as ice/water or less familiar such as a bismuth alloy.

However, in many situations, the techniques described above would be undesirable or impractical due to various reasons. The passive techniques might have a limited lifetime that is insufficient for the application, while the active techniques require additional power and subsystems. Also, for some